Case Study 18: Fuel Consumption and Route Comparison

1. Problem Statement

Compare multiple alternative driving routes based on total distance, average vehicle speed, fuel efficiency, and fuel price to calculate total fuel consumption, expenditure, and estimated travel duration to identify the most cost-effective and time-efficient route.

2. Objectives

- Evaluate route metrics considering distance, expected traffic speed, and vehicle efficiency.
- Compute fuel volume requirements and associated financial cost for each travel path.
- Compare tradeoffs between monetary cost and overall travel duration.

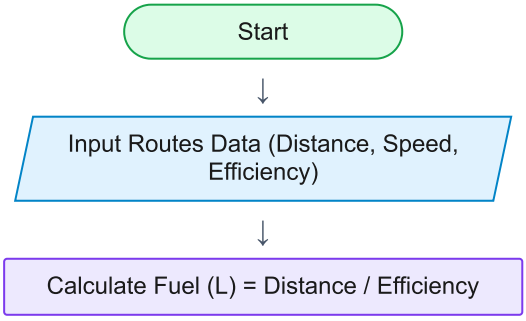
3. Assumptions

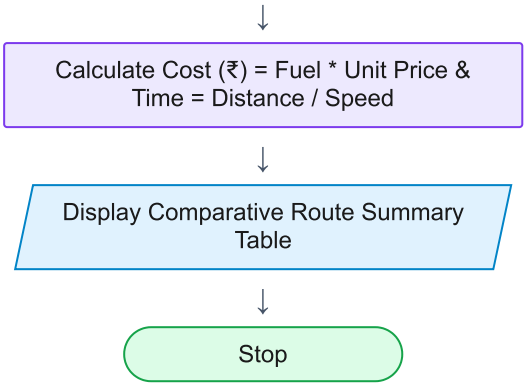
- Fuel price is fixed at **₹100.00 per Liter**.
- Vehicle fuel efficiency (km/L) varies per route based on road conditions and traffic flow.
- Speed remains uniform along each evaluated route segment.

4. Algorithm

1. Start.
2. Set `fuel_price = 100.0`.
3. Define dictionary of routes containing distance, speed, and fuel efficiency.
4. For each route option:
   - Calculate `fuel_needed = distance / efficiency`.
   - Calculate `total_cost = fuel_needed * fuel_price`.
   - Calculate `travel_time = distance / speed`.
5. Display comparison table showing fuel required, total cost, and travel duration.
6. Identify the optimal route for cost and time efficiency.
7. Stop.

5. Flowchart





6. Pseudocode

```
START fuel_price = 100.0 INPUT routes_map FOR EACH route IN routes_map DO dist =
route.distance speed = route.speed eff = route.efficiency fuel = dist / eff cost =
fuel * fuel_price time_hrs = dist / speed PRINT route.name, fuel, cost, time_hrs
ENDFOR STOP
```

7. Test Case and Expected Output

Evaluation Results across 3 Alternative Routes (Fuel Price = ₹100/L):

```
Route | Fuel (L) | Cost (₹) | Time (hrs) ----------------------------------------
------------------ Route A (Highway) | 8.0 | 800.0 | 1.5 Route B (City/Short) | 9.0
| 900.0 | 2.25 Route C (Bypass) | 8.4 | 840.0 | 1.75
```

8. Conclusion

The Fuel Consumption and Route Comparison tool equips commuters and logistics operators to balance trade-offs between travel duration and total fuel expenditure, identifying Highway Route A as both the fastest and most cost-effective option.